



WorkWithHelmetOnly

Abdullah Abo El-Hija, Zenab Abd Elghani

Advisor: Dr. Zeev Frenkel

GitHub : <https://github.com/AbdullahAboElHija/FinalProject>

L 25-1-D-25

The Proplem:

Construction sites face critical safety challenges with PPE compliance monitoring. Manual inspections are inefficient, prone to human error, and cannot provide real-time violation detection. This leads to increased workplace accidents and regulatory non-compliance.

Our Solution:

WorkWithHelmetOnly provides automated PPE detection using advanced computer vision and deep learning. The system offers real-time monitoring, instant violation alerts, comprehensive tracking across video frames, and detailed compliance reporting for construction safety management.

True Positive & True Negative



Unclear:



Architecture:



False Positive :



False Negative :



Technology Stack :

YOLOv11: Advanced object detection framework
OpenCV: Computer vision processing library
SORT: Multi-object tracking algorithm
Streamlit: Web application framework
Python: Core development language
SMTP: Email notification protocol
CSV: Data logging and export format

Performance Validation

Module	Function Tested	Results
Object Detection	Helmet Detection Accuracy	96.2% precision with 93.8% recall
PPE Recognition	Safety Vest Classification	94.8% accuracy across varied conditions
Person Tracking	Multi-object SORT Algorithm	89.4% MOTA with minimal ID switches
Alert System	Notification Sent After 3-Seconds Detection	Notification email sent successfully after 3 seconds of detection.
Database Logging	Data Storage and Retrieval	100% transaction success rate
Web Interface	User Experience Testing	4.8/5 usability score from testers
Email Notifications	Alert Delivery System	100% successful delivery rate

Dataset :

PPE Dataset (Roboflow) – 1,613 images, JPG/PNG. 8 classes.
Split: 70 % train / 20 % val / 10 % test.

